ZERO SPEED FOR EXCITED RANDOM WALKS ON THE EVENT OF NON-TRANSIENCE

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ABSTRACT. An excited random walk on $\mathbb{Z}^d$ moves, on its i-th visit to a site x, according to the i-th entry of a stack of transition probabilities attached to x; the stacks are independent and identically distributed over the sites. Kosygina and Zerner asked whether $X_n/n \to 0$ almost surely on the event that $|X_n|$ does not tend to infinity, assuming only that the stacks are i.i.d. and every entry is positive. We prove that it does, in every dimension and with no further restriction on the stacks, which may be infinite and may push in any direction. We also prove the excited-walk form of Zerner's law of large numbers for random walk in random environment: if, for a coordinate direction e, the walk tends to $+\infty$ and to $-\infty$ in direction e with probability zero, then $X_n \cdot e/n \to 0$ almost surely. The strong Markov restart in Zerner's argument is replaced by a sum over the finitely many paths the walk can follow between its first and last visits to a level.

1. INTRODUCTION

Let $E = \{\pm e_1, \dots, \pm e_d\}$ be the unit vectors of $\mathbb{Z}^d$. A *cookie environment* is a family $\omega = (\omega(x,a,i))_{x\in\mathbb{Z}^d,\, a\in E,\, i\geq 1}$ with $\omega(x,a,i) \geq 0$ and $\sum_{a\in E}\omega(x,a,i) = 1$. The sequence $\omega(x,\cdot,\cdot)$ is the *stack* at x. Given ω and $y \in \mathbb{Z}^d$, the excited random walk $(X_n)_{n\geq 0}$ starts at $X_0 = y$ and moves by

$$P_{y,\omega}\big(X_{n+1} = X_n + a \,\big|\, X_0, \dots, X_n\big) = \omega\big(X_n, a, \#\{0 \leq m \leq n : X_m = X_n\}\big), \qquad a \in E, \tag{1}$$

so on its i-th visit to a site the walk uses the i-th entry of that site's stack. The environment is random with law $\mathbb{P}$, and $P_y = \mathbb{E}[P_{y,\omega}]$ is the annealed law, where $\mathbb{E}$ is expectation with respect to $\mathbb{P}$; E_y is expectation with respect to P_y. We use the two assumptions of the survey of Kosygina and Zerner [6]:

(IID) the stacks $\omega(x,\cdot,\cdot)$, $x \in \mathbb{Z}^d$, are independent and identically distributed under $\mathbb{P}$;
(EL) $\mathbb{P}$-almost surely, $\omega(x,a,i) > 0$ for all x, a and i.

Nothing else is assumed. There is no uniform lower bound in (EL), the entries of one stack need not be independent, and a stack may have infinitely many entries that differ from the simple random walk step, with drifts of either sign.

For $\ell \in \mathbb{R}^d \setminus \{0\}$ let $A_\ell = \{X_n \cdot \ell \to +\infty\}$, and let

$$N = \big\{|X_n| \not\to \infty\big\}$$

be the event that the walk does not go to infinity, which we call non-transience. Kosygina and Zerner asked the following [6, Problem 4.7].

Problem. Assume (IID) and (EL), and let $\ell \in \mathbb{R}^d \setminus \{0\}$. Does $X_n \cdot \ell/n$ converge to 0 P_0-almost surely on N?

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For random walk in an i.i.d. elliptic random environment, which is the case in which $\omega(x, a, i)$ does not depend on i, Zerner [9, Theorem 1] proved the following: if $e \in E$ and $P_0(A_e \cup A_{-e}) = 0$, then $X_n \cdot e/n \to 0$ P_0-almost surely. In dimension one, Amir, Berger and Orenshtein [1, Theorem 1.3] proved that X_n/n converges almost surely to a deterministic limit for stationary ergodic elliptic stacks on $\mathbb{Z}$, infinite stacks included; when $P_0(N) > 0$ the limit is zero, since on N the walk visits some site infinitely often. We answer the problem in every dimension.

Theorem 1.1. *Let $d \geq 1$ and assume (IID) and (EL). Then $X_n/n \to 0$ P_0-almost surely on N. If $P_0(N) > 0$, then $X_n/n \to 0$ P_0-almost surely.*

The main step is the following one-sided statement, which holds for every coordinate direction separately.

Proposition 1.2. *Assume (IID) and (EL), and let $e \in E$. If $P_0(A_e) = 0$, then $\limsup_n X_n \cdot e/n \leq 0$ P_0-almost surely.*

Applied to e and to $-e$, Proposition 1.2 gives the excited-walk form of Zerner's theorem.

Corollary 1.3. *Assume (IID) and (EL), and let $e \in E$. If $P_0(A_e \cup A_{-e}) = 0$, then $X_n \cdot e/n \to 0$ P_0-almost surely.*

When the stacks are constant in i this is [9, Theorem 1]. When $P_0(N) > 0$, the zero-one law [6, Theorem 3.16] gives $P_0(A_e \cup A_{-e}) = 0$ for every $e \in E$, and Theorem 1.1 follows; the deduction is in Section 5.

The idea of the proof. Fix $e \in E$, call $\{x : x \cdot e = m\}$ level m, let T_m be the first time the walk is at level m, and for an integer $L \geq 1$ let $h_{m,L}$ be the time from T_m to the last visit to level m before T_{m+L}. Zerner's proof has two parts. By [9, Lemma 3], a statement about a single path, if $\limsup_n X_n \cdot e/n > 0$ then there is an integer c such that, for every L, the upper density

$$\limsup_{M\to\infty} \frac{1}{M+1} \#\{0 \leq m \leq M : \ h_{m,L} \leq c\}$$

of the levels m with $h_{m,L} \leq c$ is bounded below by a positive constant that does not depend on L. The second part bounds the probability that $h_{m,L} \leq c$. At its last visit to level m the walk is at one of finitely many sites z, for the r-th time with $r \leq c + 1$; the r-th visit to z is a stopping time, and by the strong Markov property the walk has law $P_{z,\omega}$ from then on. The probability that it then reaches level $m + L$ without returning to level m depends only on the environment at levels $\geq m$, which is independent of the path up to time T_m.

For an excited walk the law after the r-th visit to z is not $P_{z,\omega}$. It depends on which entries of the stacks at z, and at the other sites visited since T_m, have been used, and this is not determined by z and r. It is determined by the path of the walk from T_m to that visit. Let Π_c be the finite set of nearest-neighbour paths $p = (p_0, \ldots, p_s)$ with $p_0 = 0$, $s \leq c$ and $p_s \cdot e = 0$, and let ω^p be the environment obtained from ω by deleting from each stack the entries used along p. Bounding by one the probability that the walk follows p after T_m, we obtain under (IID) alone, for all $m \geq 0$ and $L > c$, on $\{T_m < \infty\}$,

$$P_0\big(T_{m+L} < \infty,\ h_{m,L} \leq c \,\big|\, X_0, \ldots, X_{T_m}\big) \leq \sum_{p\in\Pi_c} \mathbb{E}\Big[P_{p_s,\omega^p}\big(T_L < \infty,\ X_n \cdot e > 0 \text{ for } 1 \leq n \leq T_L\big)\Big]$$

(Lemma 3.2). The right side depends neither on m nor on the path up to time T_m. Each term decreases, as $L \to \infty$, to the expected probability that the walk in ω^p started at p_s stays above level 0 and reaches every level, and under (EL) this is zero when $P_0(A_e) = 0$ (Lemma 3.1).

Hence, almost surely, the upper density of the levels m with $h_{m,L} \le c$ is at most the right side, which tends to 0 as $L \to \infty$ for each c, and Zerner's lemma gives $\limsup_n X_n \cdot e/n \le 0$.

Related work. Excited random walk was introduced by Benjamini and Wilson [3], with one cookie of drift in direction e_1 at each site. Zerner [10] and Menshikov, Popov, Ramírez and Vachkovskaia [7] studied walks whose drifts point into a half-space, and Bérard and Ramírez [2] proved a central limit theorem for the Benjamini–Wilson walk; in these models the walk is transient and $P_0(N) = 0$. The laws of large numbers of Holmes [4] and Pham [8] in high dimensions assume simple transverse steps. Theorem 4.8 of the survey [6] proves a law of large numbers under the assumption $P_0(A_{\ell_i} \cup A_{-\ell_i}) = 1$ for a basis $\ell_1, \dots, \ell_d$, which the survey relates to Problem 4.7; Corollary 1.3 covers a coordinate direction in which this probability is 0. In dimension one, besides [1], infinite stacks were treated by Kosygina and Peterson [5] for Markovian stacks. A table of notation is in Appendix A.

2. Preliminaries

2.1. **Results from the survey.** We use three results of [6], each proved there under (IID) and (EL).

- (R1) [6, Theorem 2.5] The range $\{X_n : n \ge 0\}$ is either P_0-almost surely finite or P_0-almost surely infinite.
- (R2) [6, Lemma 3.18] If the range is P_0-almost surely infinite, then for every $\ell \in \mathbb{R}^d \setminus \{0\}$ and all real $u < w$, the walk almost surely does not stay in the half-space $\{x : x \cdot \ell \le w\}$ forever while visiting the slab $\{x : u \le x \cdot \ell \le w\}$ infinitely often.
- (R3) [6, Theorem 3.16] For every $\ell \in \mathbb{R}^d \setminus \{0\}$, $P_0(A_\ell \cup A_{-\ell}) \in \{0, 1\}$.

The survey states (R2) under (IID) and $\mathbb{E}[\omega(0, e, 1)] > 0$ for all $e \in E$, which (EL) implies. We use (R2) only with ℓ a coordinate vector and u, w integers.

Lemma 2.1. *Assume (IID) and (EL), suppose the range is P_0-almost surely infinite, and let $e \in E$. Then*

$$P_0\Big(\inf_n X_n \cdot e > -\infty,\ X_n \cdot e \not\to +\infty\Big) = 0.$$

Proof. On the event in question there are integers $a < b$ such that $X_n \cdot e \ge a$ for all n and $X_n \cdot e \le b$ for infinitely many n. For fixed $a < b$ this is the event of (R2) with $\ell = -e$, $w = -a$ and $u = -b$, which is null. Take the union over the countably many pairs (a, b). □

2.2. **The counting lemma.** The quantities T_m and $h_{m,L}$ of Section 1 are defined for every nearest-neighbour path $(x_n)_{n \ge 0}$ in $\mathbb{Z}^d$ with $x_0 \cdot e = 0$, a direction $e \in E$ and integers $m \ge 0$, $L \ge 1$; in formulas,

$$T_m = \inf\{n \ge 0 : x_n \cdot e = m\}, \qquad I_{m,L} = \{n : T_m \le n < T_{m+L},\ x_n \cdot e = m\},$$

and $h_{m,L} = \max I_{m,L} - T_m$ when $T_{m+L} < \infty$, with the convention $h_{m,L} = \infty$ otherwise. For an integer $c \ge 0$ put

$$F_{M,L}(c) = \frac{1}{M+1} \sum_{m=0}^{M} \mathbf{1}\{h_{m,L} \le c\}.$$

Since $h_{m,L}$ does not decrease in L, $F_{M,L}(c)$ does not increase in L. The next lemma is the deterministic part of Zerner's argument; it concerns a single path and its proof in [9] uses no property of the environment.

Lemma 2.2 ([9, Lemma 3])**.** *If $\limsup_n x_n \cdot e/n > 0$, then there is an integer $c \ge 0$ with*

$$\inf_{L \ge 1} \limsup_{M \to \infty} F_{M,L}(c) > 0.$$

3. The estimate at a new level

Fix $e \in E$ and recall Π_c and ω^p from Section 1; in formulas, for $p \in \Pi_c$ let $k_p(x) = \#\{0 \le t < s : p_t = x\}$ be the number of departures from x along p, and $\omega^p(x,a,i) = \omega(x,a,i+k_p(x))$. The endpoint p_s has not been left along p after its last visit, so its next entry is not removed. For $L > c$ define

$$q_L^p(\omega) = P_{p_s,\omega^p}\big(T_L < D\big), \qquad D = \inf\{n \ge 1 : X_n \cdot e \le 0\},$$

for the walk started at p_s. On $\{T_L < D\}$ the walk leaves level 0 at its first step and stays strictly above it until it reaches level L, so q_L^p depends only on the stacks at sites x with $x \cdot e \ge 0$, even if p itself visits negative levels. As L increases, q_L^p decreases to

$$q_\infty^p(\omega) = P_{p_s,\omega^p}\Big(X_n \cdot e > 0 \text{ for all } n \ge 1 \text{ and } \sup_n X_n \cdot e = \infty\Big),$$

the probability of reaching every level without returning to level 0. Put

$$a_L(c) = \sum_{p \in \Pi_c} \mathbb{E}\big[q_L^p\big], \qquad L > c.$$

Lemma 3.1. *Assume (IID) and (EL), and suppose $P_0(A_e) = 0$. Then for every $c \ge 0$, $a_L(c) \to 0$ as $L \to \infty$.*

Proof. Fix $p \in \Pi_c$ and let $w_p(\omega) = P_{0,\omega}\big((X_0,\dots,X_s) = p\big)$, a finite product of entries of ω, which is positive $\mathbb{P}$-almost surely by (EL). After following p the walk sits at p_s with exactly the entries ω^p left, so by (1), $w_p\, q_\infty^p$ is the probability under $P_{0,\omega}$ that the walk follows p and afterwards stays strictly above level 0 and reaches every level. If the range is P_0-almost surely finite, this event is P_0-null. Otherwise, since $X_n \cdot e \ge -c$ for all n on this event, Lemma 2.1 shows that it is contained in A_e up to a P_0-null set. In both cases $\mathbb{E}[w_p q_\infty^p] \le P_0(A_e) = 0$, and since $w_p > 0$ almost surely, $q_\infty^p = 0$ almost surely. By dominated convergence $\mathbb{E}[q_L^p] \to 0$, and Π_c is finite. □

Lemma 3.2. *Assume (IID). Let $m \ge 0$, $c \ge 0$ and $L > c$. Let $\mathcal{F}_n = \sigma(X_0,\dots,X_n)$ and $\mathcal{F}_\infty = \sigma(X_n : n \ge 0)$; for a stopping time T, $\mathcal{F}_T$ is the σ-field of events determined by the path up to time T. Let*

$$B_{m,L,c} = \{T_{m+L} < \infty,\ h_{m,L} \le c\}.$$

Then $P_0(B_{m,L,c} \mid \mathcal{F}_{T_m}) \le a_L(c)$ on $\{T_m < \infty\}$.

Proof. Fix a finite path $\eta = (\eta_0,\dots,\eta_t)$ with $\eta_0 = 0$ whose first visit to level m is at step t, let $x = \eta_t$, and let $w_\eta(\omega) = P_{0,\omega}\big((X_0,\dots,X_t) = \eta\big)$. Before time t the walk leaves only sites at levels below m, so w_η is a product of entries of stacks at sites y with $y \cdot e < m$.

On $B_{m,L,c} \cap \{(X_0,\dots,X_t) = \eta\}$ let $t+s$ be the last visit to level m before T_{m+L}, so that $s = h_{m,L} \le c$, and let $p = (X_{t+r} - x)_{0 \le r \le s}$; then $p \in \Pi_c$. Conversely, if the walk follows η, then $x + p$ for some $p \in \Pi_c$, and then reaches level $m+L$ before returning to level m, it lies in $B_{m,L,c}$ with $h_{m,L} = s$, because the path $x+p$ has at most $c < L$ steps and does not reach level $m+L$. These events are disjoint for different p: if p is a proper initial piece of p', a walk that follows $x + p'$ is at level m at a step after $t+s$ and before T_{m+L}. After following η and then $x+p$, the walk sits at $x + p_s$, and by (1) the probability that it then reaches level $m+L$ before returning to level m is $q_L^p(\theta_x\omega')$, where ω' is the environment with the entries used by η removed and $(\theta_x\omega')(z,a,i) = \omega'(x+z,a,i)$. As noted above, q_L^p reads only stacks at levels $\ge m$ after the shift, and these were not touched by η, so $q_L^p(\theta_x\omega') = q_L^p(\theta_x\omega)$. The probability

of following η and then $x+p$ is at most $w_\eta(\omega)$, so

$$P_{0,\omega}\big(B_{m,L,c},\ (X_0,\dots,X_t)=\eta\big) \le w_\eta(\omega)\sum_{p\in\Pi_c} q_L^p(\theta_x\omega).$$

The factor w_η depends only on the stacks at levels $< m$ and the sum only on the stacks at levels $\ge m$, which are independent by (IID); by translation invariance $\mathbb{E}[q_L^p\circ\theta_x]=\mathbb{E}[q_L^p]$. Taking expectations, $P_0\big(B_{m,L,c},\ (X_0,\dots,X_t)=\eta\big) \le a_L(c)\,P_0\big((X_0,\dots,X_t)=\eta\big)$. An event of $\mathcal{F}_{T_m}$ contained in $\{T_m<\infty\}$ is a disjoint union of events $\{(X_0,\dots,X_t)=\eta\}$ of this kind, and summing over them gives the claim. □

4. Proof of Proposition 1.2

If the range is P_0-almost surely finite, then X_n is bounded and $X_n/n\to 0$, so by (R1) we may assume that it is almost surely infinite. Suppose $P_0(A_e)=0$.

Fix $c\ge 0$, $L>c$ and a residue $r\in\{0,\dots,L-1\}$, and put $m_k=r+kL$, $\mathcal{G}_k=\mathcal{F}_{T_{m_k}}$ and $Z_k=\mathbf{1}_{B_{m_k,L,c}}$. The event $B_{m_k,L,c}$ is determined by the path up to $T_{m_k+L}=T_{m_{k+1}}$, so Z_k is $\mathcal{G}_{k+1}$-measurable, and $Z_k-E_0[Z_k\mid\mathcal{G}_k]$ is a bounded martingale difference sequence. By the strong law of large numbers for such sequences and Lemma 3.2,

$$\limsup_{K\to\infty}\frac{1}{K}\sum_{k<K} Z_k \le a_L(c) \qquad P_0\text{-a.s.}$$

If some T_{m_k} is infinite then $Z_j=0$ for all $j\ge k$ and the bound holds trivially. Averaging over the L residues and intersecting over the countably many pairs (c,L), almost surely, for all $c\ge 0$ and $L>c$,

$$\limsup_{M\to\infty}\frac{1}{M+1}\sum_{m=0}^{M}\mathbf{1}_{B_{m,L,c}} \le a_L(c). \tag{2}$$

Now take a path in the almost sure event of (2) with $\limsup_n X_n\cdot e/n>0$. Since $h_{m,L}=\infty$ when $T_{m+L}=\infty$, $B_{m,L,c}=\{h_{m,L}\le c\}$, and the left side of (2) is $\limsup_M F_{M,L}(c)$. By Lemma 3.1, $\inf_{L>c}\limsup_M F_{M,L}(c)=0$ for every c; since $F_{M,L}(c)$ does not increase in L, the infimum over all $L\ge 1$ is also zero. This contradicts Lemma 2.2. Hence $\limsup_n X_n\cdot e/n\le 0$ almost surely. □

5. Proofs of Corollary 1.3 and Theorem 1.1

Proof of Corollary 1.3. If $P_0(A_e\cup A_{-e})=0$, then $P_0(A_e)=P_0(A_{-e})=0$. Proposition 1.2, applied to e and to $-e$, gives $\limsup_n X_n\cdot e/n\le 0$ and $\liminf_n X_n\cdot e/n\ge 0$ almost surely. □

Proof of Theorem 1.1. If $P_0(N)=0$ there is nothing to prove. Suppose $P_0(N)>0$ and let $j\in\{1,\dots,d\}$. On $A_{e_j}\cup A_{-e_j}$ we have $|X_n|\to\infty$, so N is disjoint from it and $P_0(A_{e_j}\cup A_{-e_j})<1$. By (R3) this probability is 0, and Corollary 1.3 gives $X_n\cdot e_j/n\to 0$ almost surely. Hence $X_n/n\to 0$ P_0-almost surely, and in particular on N. □

Remark 5.1. The event N need not be the event that every site is visited infinitely often. For instance, in $d=2$ give each site an independent uniformly chosen $a\in E$, and let the i-th entry of its stack put mass $1-3\varepsilon_i$ on a and $\varepsilon_i=2^{-i-4}$ on each other direction. With positive probability the directions at 0 and e_1 point at each other and the walk moves back and forth between these two sites forever, since

$$\frac{1}{16}\prod_{i\ge 1}\big(1-3\cdot 2^{-i-4}\big)^2>0.$$

This environment satisfies (IID) and (EL), and on that event the speed is zero, as Theorem 1.1 requires.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

A Codex agent (OpenAI) proved Theorem 1.1; a second Codex agent checked the proof as a referee; a Claude agent (Anthropic) checked the steps and prepared the manuscript. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Code availability

Exact finite-horizon checks of the decomposition in Lemma 3.2, checks of the averages in Lemma 2.2 on random paths, and a simulation of the example in Remark 5.1 are in the repository

`https://github.com/yspennstate/excited-walk-zero-speed-non-transience`.

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Appendix A. Notation

Symbol	Meaning	First appears
E	the $2d$ unit vectors of $\mathbb{Z}^d$	Section 1
$\omega(x,a,i)$	probability of step a on the i-th visit to x (an entry of the stack at x)	Section 1
$P_{y,\omega}$, P_y, $\mathbb{P}$	quenched law, annealed law, law of the environment	(1)
$\mathbb{E}$, E_y	expectation with respect to $\mathbb{P}$ and to P_y	Section 1
A_ℓ	the event $X_n \cdot \ell \to +\infty$	Section 1
N	the event $\lvert X_n\rvert \not\to \infty$	Section 1
level m	the set $\{x : x \cdot e = m\}$	Section 1
T_m, $h_{m,L}$	first time at level m; time from T_m to the last visit to level m before T_{m+L}	Section 1
Π_c, ω^p	short paths; the environment with the entries used by p removed	Section 1
$I_{m,L}$	visits to level m before level $m+L$	Section 2
$F_{M,L}(c)$	fraction of levels $m \le M$ with $h_{m,L} \le c$	Section 2
k_p	departure counts of p	Section 3
q_L^p, q_∞^p, D	chance of reaching level L before returning to level 0, its limit, the return time	Section 3
$a_L(c)$	$\sum_{p\in\Pi_c} \mathbb{E}[q_L^p]$	Section 3
w_p	quenched probability of following p	Lemma 3.1
$B_{m,L,c}$	the event $T_{m+L} < \infty$, $h_{m,L} \le c$	Lemma 3.2
$\mathcal{F}_n$, $\mathcal{F}_\infty$, $\mathcal{F}_T$	the natural filtration, its limit, the σ-field at a stopping time	Lemma 3.2
w_η, ω', θ_x	quenched probability of following η; the environment with the entries used by η removed; the shift by x	Lemma 3.2

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